East West University
Department of Computer Science and Engineering
Course: CSE464
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Lab 02: Implementing Loops and Conditional Statements in PL/SQL

Lab Objectives:

- Understand the use of control structures in PL/SQL.
- Implement iterative logic using different loop constructs (FOR, WHILE, LOOP).
- Apply conditional logic using IF-THEN-ELSE and CASE statements.

Topics Covered:

- FOR, WHILE, and basic LOOP constructs
- Conditional structures:
  - IF-THEN
  - IF-THEN-ELSE
  - IF-THEN-ELSIF
  - CASE statement
- Combining loops and conditionals in a single PL/SQL block

Lab Outcomes:

By the end of this lab session, students were able to:

1. **Implement Loops** – Write iterative logic using FOR, WHILE, and LOOP constructs in PL/SQL.
2. **Apply Conditional Statements** – Develop decision-making logic using IF-THEN-ELSE and CASE statements.
3. **Structure PL/SQL Code Effectively** – Combine loops and conditionals to write complete, logical, and well-structured PL/SQL blocks.

Lab Activities

1. FOR Loop

Syntax:

```
FOR counter IN [start]..[end] LOOP
   -- statements
END LOOP;
```

Example:

```
BEGIN
  FOR i IN 1..10 LOOP
    DBMS_OUTPUT.PUT_LINE('Number: ' || i);
  END LOOP;
END;
```

2. WHILE Loop

Syntax:

```
WHILE condition LOOP
   -- statements
END LOOP;
```

Example:

```
DECLARE
  i NUMBER := 1;
BEGIN
  WHILE i <= 5 LOOP
    DBMS_OUTPUT.PUT_LINE('iteration: ' || i);
    i := i + 1;
  END LOOP;
END;
```

3. Basic LOOP

Syntax:

```
LOOP
   -- statements
   EXIT WHEN condition;
END LOOP;
```

Example:

```
DECLARE
  i NUMBER := 2;
BEGIN
  LOOP
    EXIT WHEN i > 20;
    DBMS_OUTPUT.PUT_LINE(i);
    i := i + 2;
  END LOOP;
END;
```

4. IF-THEN-ELSE **Statement**

Syntax:

```
IF condition THEN
   -- statements
ELSIF condition THEN
   -- statements
ELSE
   -- statements
END IF;
```

Example:

```
DECLARE
  num NUMBER := -5;
BEGIN
  IF num > 0 THEN
    DBMS_OUTPUT.PUT_LINE('Positive');
  ELSIF num < 0 THEN
    DBMS_OUTPUT.PUT_LINE('Negative');
  ELSE
    DBMS_OUTPUT.PUT_LINE('Zero');
  END IF;
END;
```

5. CASE **Statement**

Syntax:

```
CASE variable
  WHEN value1 THEN
     -- statements
  WHEN value2 THEN
     -- statements
  ELSE
     -- statements
END CASE;
```

Example:

```
DECLARE
    day VARCHAR2(30);
BEGIN
    day := '&day';
    CASE day
        WHEN 'sunday' THEN
            DBMS_OUTPUT.PUT_LINE('class');
        WHEN 'tuesday' THEN
            DBMS_OUTPUT.PUT_LINE('lab');
        ELSE
            DBMS_OUTPUT.PUT_LINE('there is no class');
    END CASE;
END;
```

Practice Problems – PL/SQL Control Structures

1. Sum of First N Natural Numbers

Write a PL/SQL block that reads a number N from the user and calculates the sum of the first N natural numbers using a LOOP.

2. Factorial Calculation

Write a PL/SQL block that reads a number N from the user and calculates the factorial of N using a FOR loop.

3. Even or Odd Number Check

Write a PL/SQL block that takes a number as input and checks whether it is even or odd using an IF-THEN-ELSE statement.

4. Student Grade Calculation (IF-THEN-ELSIF)

Write a PL/SQL block that accepts a student's score and assigns a grade based on the following conditions:

- 90–100: Grade A
- 80–89: Grade B
- 70–79: Grade C
- Below 70: Fail

Use an IF-THEN-ELSIF statement to implement this logic.

5. Student Grade Calculation (CASE Statement)

Repeat the previous problem using a CASE statement instead of IF-THEN-ELSIF.

6. Find the Largest of Three Numbers

Write a PL/SQL block that reads three numbers and determines the largest using an IF-THEN-ELSE statement.

7. Count Digits in a Number

Write a PL/SQL block that takes a number as input and counts the total number of digits using a WHILE loop.